A Study on "Vision Transformers Need Registers"

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“Vision Transformers Need Registers”: Overview

Darcet et al. 2024, *ICLR*, Vision Transformers Need Registers

- **Paper Link:** https://openreview.net/forum?id=2dnO3LLiJ1
- **Repo. Link:** https://github.com/kyegomez/Vit-RGTS (for replication)
- **TL;DR:**
  - Vision Transformers (ViTs) exhibit **artifacts**:

    high-norm tokens in low-information regions

  - In both supervised and self-supervised; disrupt downstream tasks.
  - **Registers**: additional tokens; only for internal computation.
  - This approach:
    - Achieves **SOTA** on dense prediction tasks.
    - Enables **object discovery** in large models.
    - Produces smoother **feature and attention maps**.

1. **Problems**
    - **Motivation**
    - **Analysis**
2. Method
3. Experiments
4. Conclusion
5. Further work

Motivation

Big picture:

$\boxed{\text{Embedding images into generic features that can serve multiple purposes in CV}}$

- **Earlier, Handcrafted principles:** e.g., SIFT by Lowe 2004, IJCV.
- **Today, “Labels are expensive”**:**Pretrain** $\rightarrow$ **feature extractor**
  - For tasks with abundant data, pretrain a model.
  - Use parts of the model as a feature extractor for downstream tasks.

Motivation (cont.)

To bring us on the same page:

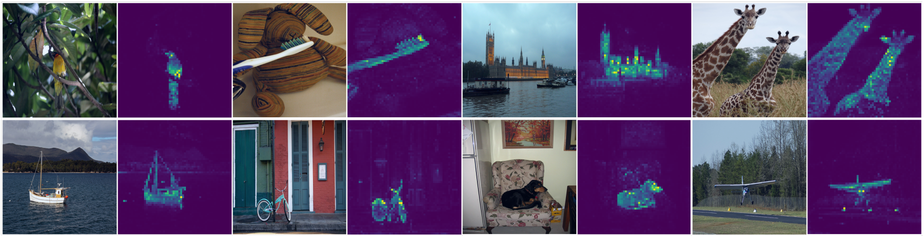


Figure 1: Self-attention from a ViT trained with no supervision (Caron et al. 2021).

- Object segmentations from the self-attention[1].

[1] Self-attention of the [CLS] query produced by the last attention layer (**attention map**).

Motivation (cont.)

The paper's motivation:

Observed: In LOST, DINOv2 *worse* than DINO. Analyze, fix these ViTs.

LOST[2]: DINO features for Obj. detection[3]. → A clean attention map?.

[2] Siméoni et al. 2021, *BMVC*; Seed points → full object regions.

[3] Not using attention maps (DINO-seg), but patch features.

Motivation (cont.)

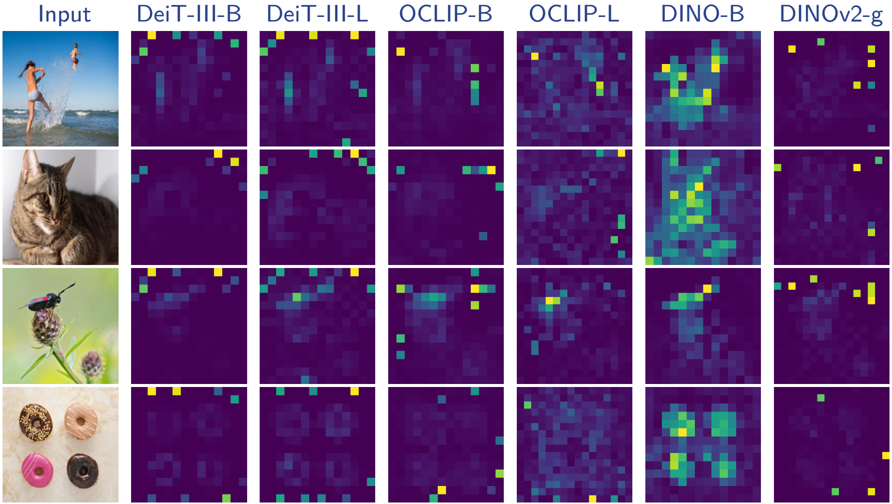


Figure 2: *All models but DINO* exhibit peaky outlier values in the attention maps (Darcet et al. 2024).

Artifacts are high-norm outlier tokens

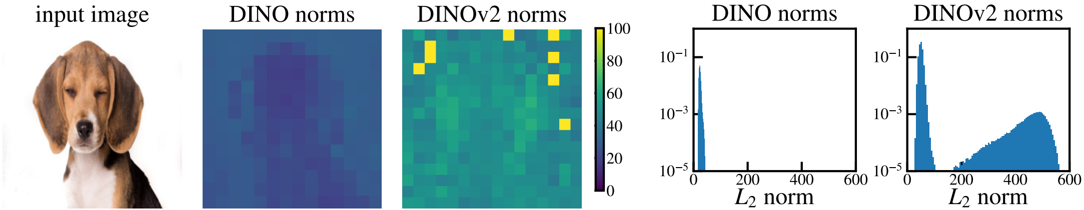
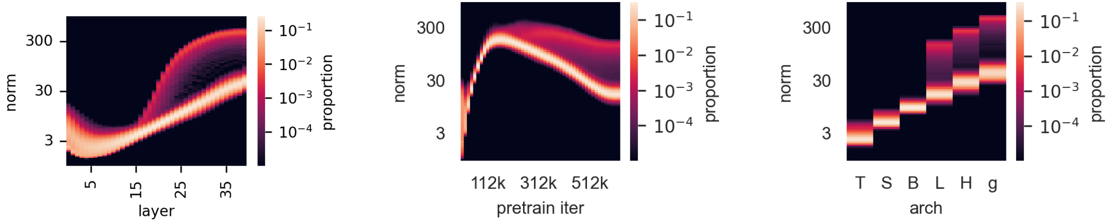


Figure 3: Comparison of local feature norms (Darcet et al. 2024)[4]

[4] 150 is the threshold used for artifacts in the paper if not otherwise specified.

Analysis (cont.)

Outliers appear during the training of large models



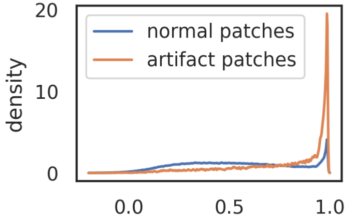
(a) Norms along layers. (b) Norms along iterations. (c) Norms across model size.

Figure 4: Illustration of several properties of outlier tokens (Darcet et al. 2024)[5].

[5] 40-layer DINOv2 ViT-g model.

Analysis (cont.)

High-norm tokens appear where patch information is redundant (Fig. 5a)



(a) Cosine similarity to neighbors.

	position prediction		reconstruction
	top-1 acc	avg. distance ↓	L2 error ↓
normal	**41.7**	**0.79**	**18.38**
outlier	22.8	5.09	25.23

(b) Linear probing for local information.

Figure 5: (a): Distribution of cosine similarity between input patches and their 4 neighbors. (b): Local information probing on normal and outlier patch tokens (Darcet et al. 2024).

Analysis (cont.)

High-norm tokens hold little local information (Tab. 5b)

Artifacts hold global information (Tab. 1)

Table 1: Image classification via linear probing on normal and outlier patch tokens (Darcet et al. 2024)[6].

	IN1k	P205	Airc.	CF10	CF100	CUB	Cal101	Cars	DTD	Flow.	Food	Pets	SUN	VOC
`[[CLS]]`	**86.0**	**66.4**	**87.3**	**99.4**	**94.5**	**91.3**	<u>96.9</u>	**91.5**	**85.2**	**99.7**	**94.7**	**96.9**	**78.6**	<u>89.1</u>
normal	65.8	53.1	17.1	97.1	81.3	18.6	73.2	10.8	63.1	59.5	74.2	47.8	37.7	70.8
outlier	<u>69.0</u>	<u>55.1</u>	<u>79.1</u>	<u>99.3</u>	<u>93.7</u>	<u>84.9</u>	**97.6**	<u>85.2</u>	<u>84.9</u>	<u>99.6</u>	<u>93.5</u>	<u>94.1</u>	<u>78.5</u>	**89.7**

[6] Also report the accuracy of classifiers learnt on the class token.

1. Problems
2. **Method**
   - **Hypothesis**
   - **Remediation**
3. Experiments
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Hypothesis

In ViT

- Large, long iterations $\rightarrow$ artifacts.
- *Artifacts are not inherently bad* [7].
- However, LOST suffers [8].

[7] They often appear in low-information regions and their local role can be recovered by surrounding tokens.

[8] And other tasks that rely on *clean local features*.

Hypothesis (cont.)

Inspired by Bulatov et al. 2022, add *registers* tokens[9]:

$\boxed{\text{Additional learnable input tokens, discarded in output.}}$

This should isolate the global context, preserving clean, local patch features.

[9] First proposed for remediate translation in NLP.

Remediation

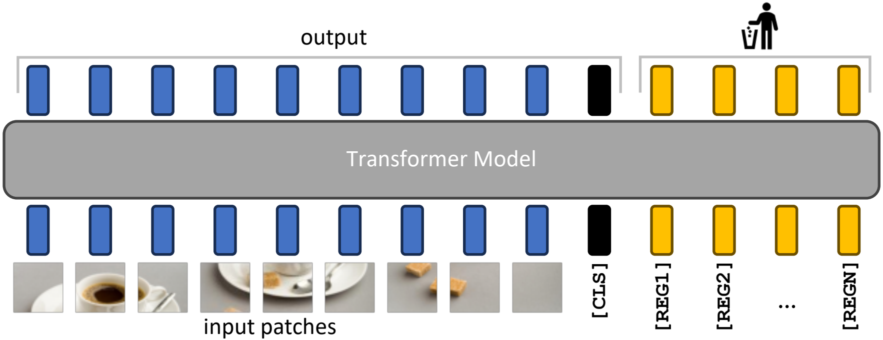

Figure 6: Illustration of the proposed remediation and resulting model. (Darcet et al. 2024).

1. Problems
2. Method
3. **Experiments**
   - **Evaluation**
   - **Ablation study**
   - **Other experiments**
4. Conclusion
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Experimental Setup

Table 2: Summary of models used in this paper's experiments.

Method	Sup.	Dataset	Arch.	Paper
DEIT-III [10]	Label	ImageNet-22k	ViT-B	Touvron et al. 2022
OpenCLIP [11]	Text	Shutterstock	ViT-B/16	Ilharco et al. 2021
DINOv2 [12]	Self	ImageNet-22k	ViT-L	Oquab et al. 2024

[10] https://github.com/facebookresearch/deit

[11] https://github.com/mlfoundations/open_clip

[12] https://github.com/facebookresearch/dinov2

Evaluation

Register tokens effectively remove the norm outliers that were present previously.

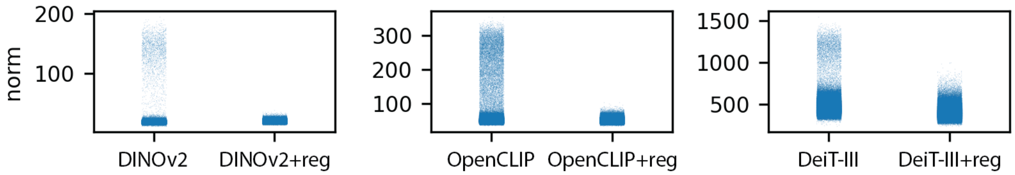

Figure 7: Effect of register tokens on the distribution of output norms (Darcet et al. 2024).

Evaluation (cont.)

Using registers not only *does not degrade performance*, but even:

$\boxed{\text{improves performance by a slight margin in some cases (Tab. 3)}}$

Evaluation (cont.)

Table 3: Evaluation of downstream performance of the models that we trained, with and without registers.

	ImageNet Top-1	ADE20k mIoU	NYUd rmse ↓
DeiT-III	84.7	38.9	0.511
DeiT-III+reg	84.7	39.1	0.512
OpenCLIP	78.2	26.6	0.702
OpenCLIP+reg	78.1	26.7	0.661
DINOv2	84.3	46.6	0.378
DINOv2+reg	84.8	47.9	0.366

(a) Linear evaluation with frozen features.

	ImageNet Top-1
OpenCLIP	59.9
OpenCLIP+reg	60.1

(b) Zero-shot classification.

Ablation study

To remove artifacts[13]::

$\boxed{\text{One register is sufficient, more leads to improved downstream performance. (Fig. 8)}}$

[13] Evaluated on three tasks (ImageNet, ADE-20k and NYUd)

Ablation study (cont.)

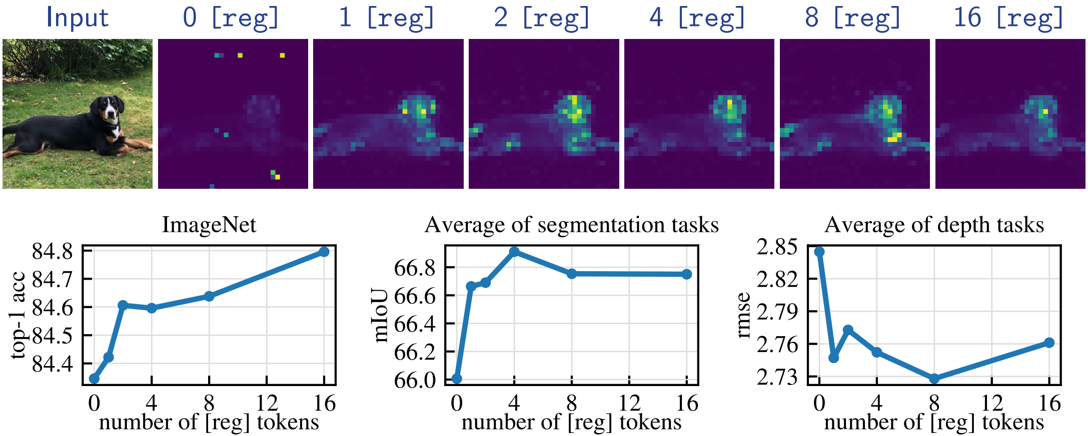

Figure 8: Ablation of the the number of register tokens used with a DINOv2 model (Darcet et al. 2024).

Other experiments - Object discovery

Registers *significantly* improve object discovery performance (Tab. 4)

	VOC 2007	VOC 2012	COCO 20k
DeiT-III	11.7	13.1	10.7
DeiT-III+reg	27.1	32.7	25.1
OpenCLIP	38.8	44.3	31.0
OpenCLIP+reg	37.1	42.0	27.9
DINOv2	35.3	40.2	26.9
DINOv2+reg	55.4	60.0	42.0

Table 4: Unsupervised Object Discovery using LOST on models with and without registers (Darcet et al. 2024)[14].

[14] Performance measured using corloc.

Other experiments - Qualitative evaluation

Similarly to slot attention (Locatello et al. 2020)[15]:

Register tokens attend to different parts of the feature map.

(It's a pity that this behaviour is not explored further in the paper.)

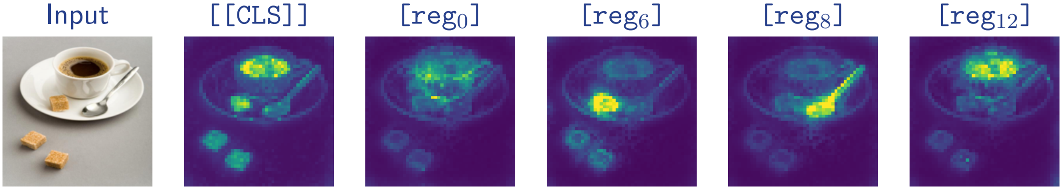

Figure 9: Comparison of the attention maps of the [CLS] and register tokens (Darcet et al. 2024).

[15] This behaviour was never required from the model, and emerged naturally from training.

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Conclusion

- *Artifacts*.
- — likely due to iterations and larger model.
- A possible interpretation: the model reuse low-info areas to store *global context*.
- *Artifacts are not inherently bad*, but can hurt LOST[16].
- *Register tokens* act as a decoupling mechanism.
- Different register tokens seem to focus on *different objects*.

[16] And other tasks that rely on clean feature maps

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Further work

- *One more benchmark*. Object discovery [TBD]
  - *Open Images, Object365*, WIDER FACE, DeepFashion2, KITTI, AIOD
- *Improve the model / one more benchmark*:

Figure 10: Preliminary results of our study.

	Tiny ImageNet[17] Top-1
DeiT-B/16-D	91.0
DeiT-B/16-D+reg	90.8

(a) Linear evaluation with frozen features.

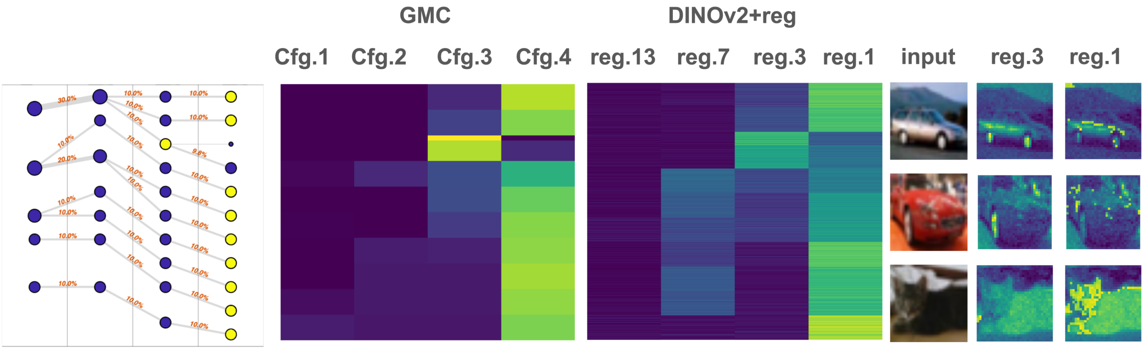
	ImageNet Top-1
OC(ResNet50)	59.6
OC(ResNet50)+reg	60.0
OC(ViT-B/32)	63.2
OC(ViT-B/32)+reg	61.4

(b) Zero-Shot Classification.

[17] https://paperswithcode.com/dataset/tiny-imagenet

Further work (cont.)

- Different registers may attend to different objects — test this more directly.
  - CLEVR, COCO Panoptic, GQA
- *Move beyond ViT*. e.g. Sequence data, matrix data. Reached T10 on a recent Kaggle[18]
- *Move beyond ViT*. e.g. Configurations from Multi-Resolution Graph-based Clustering[19]



[18] https://www.kaggle.com/competitions/stanford-rna-3d-folding

[19] https://www.qqgjyx.com/mheatmap/

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